owed_report — guide

Revision: 1 **Last modified:** 2026-09-24T19:08:01Z

Source: `constitution/scripts/codegraph/owed_report.py` (130 lines). Citations below are `owed_report.py:<line>`.

Overview

A generated, read-only report of which governed indexing tools still lack tests or a §11.4.18 companion guide (header lines 1-15). The constitution states the rule "a tool lacking its own tests or its guide MUST NOT be pointed at a live index" but does not list which tools currently lack them; a prose list would go stale, so this tool derives the answer from the file tree on every run (lines 4-7).

It is informational by default (exit 0) and becomes blocking only with `--strict` (line 126).

Prerequisites

- `python3`, standard library only (line 14, import on line 24).
- Read access to the constitution root (default: two directories above the script, line 88).

Usage

```
python3 scripts/codegraph/owed_report.py              # text table
python3 scripts/codegraph/owed_report.py --json       # machine-readable
python3 scripts/codegraph/owed_report.py --strict     # exit 1 if anything is owed
python3 scripts/codegraph/owed_report.py --root <constitution dir>
```

Flag	Meaning	Source
`--root ROOT`	Constitution root to scan (default: two levels above the script).	lines 88-89
`--json`	Print JSON instead of the table.	lines 90, 117-118
`--strict`	Exit 1 when at least one tool owes something.	lines 91, 126
`-h`, `--help`	argparse help (first line of the module docstring is the description).	line 87

Inputs

The file tree under `--root` only (nothing on stdin, no environment variables read in the source).

Tool discovery (lines 43-53, extensions line 26, skip list line 28):

Kind	Files considered
`tool`	`scripts/codegraph/*.sh`, `*.py`, `*.js`
`runner_patch`	`scripts/codegraph/runner_patches/*.py`, except `__init__.py` and `common.py`
`tool`	`scripts/lumen/*.sh`, `*.py`

Regular files only; the result is sorted by relative path (lines 38, 53).

Test attribution (lines 56-83, 100-107). A tool counts as tested when a file under `<tooldir>/tests/` (recursive; `fixtures` and `__pycache__` directories excluded, line 60; only `.sh .py .js .bats`, line 27; files that fail to read as UTF-8 are skipped, lines 68-69) either:

1. is named `test_<stem>.<ext>`, `test_<stem>_mutate.<ext>`, `test_<stem>_mutations.<ext>` or `t_<stem>.<ext>` (`name_matches`, lines 80-83); or

2. contains the tool's exact file name (runner patches: the stem, i.e. the patch id, line 106) bounded by non-identifier characters. The lookbehind also rejects `.` and `-` before the token, and a file name may not be followed by `.<alnum>` (e.g. `x.js.bak` does not match) (lines 74-77).

For runner patches the tests directory is `scripts/codegraph/tests` (`tooldir` on line 50).

Guide attribution (lines 108-112): the first existing file among `docs/scripts/<stem>.md` and `docs/scripts/<file name>.md` . Only file existence is checked (`os.path.isfile`), not the guide's content.

Outputs

`owed` for a row is `["tests"]` when no test matched, plus `["guide"]` when no guide file exists (line 113).

Text mode (lines 119-125): a header row `TOOL KIND TESTS GUIDE` , one row per tool with `yes` / `NO` columns and a trailing `OWED: tests, guide` when something is owed, then `owed: <n> of <total> tool(s)` .

`--json` (lines 117-118): `{"root", "tools", "owed_count"}` printed with `indent=1, sort_keys=True` ; each tool row has `path` , `kind` , `tested` , `tests` , `guide` , `owed` (line 114).

Exit codes

Code	Meaning	Source
0	Report produced (also when tools owe something, unless `--strict`)	line 126
1	`--strict` and at least one tool owes tests or a guide	line 126
2	argparse usage error (unknown argument)	argparse default; confirmed below
3	REFUSED: root is not a directory, or no tool was discovered	lines 31-33, 94-98

Exit 3 on an empty scan is deliberate: a blind scan must never read as "nothing owed" (header lines 11-12, refusal text on line 98).

Edge cases

- **Test evidence is textual.** Attribution by tool name (rule 2 above) searches the whole text of each test file, so a mention in a comment also counts (line 107). The tool proves a reference exists, not that the test executes the tool.
- **Guide evidence is existence only** (line 110); an empty or stale guide file clears the `guide` debt.
- **Tests are cached per tool directory** (lines 100-105), so `scripts/codegraph` and its `runner_patches` share one scan.
- `scripts/lumen` tests are read from `scripts/lumen/tests` (line 45, `read_tests(root, tooldir)` on line 105).
- The test-file header (`tests/test_owed_report.py`, line 13) and this tool's own header (line 15) cite a `test_owed_report_mutations.sh`; UNCONFIRMED whether it exists: `ls tests | grep -i owed` on the verification date listed only `test_owed_report.py`.

Internal behaviour

`main()` (line 86): parse flags, refuse on a missing root (lines 94-95), `discover()` (line 96), refuse when nothing was found (lines 97-98), then per tool compute the stem, load that directory's tests once, match, look up the guide, and append a row (lines 100-114). Nothing is ever written; the only file operations are `os.listdir`, `os.walk`, `os.path.isfile` and read-mode `open`.

Related

Guide / file	Relation
codegraph_guard	Sibling gate in the same tool set
codegraph_safe_helper	One of the tools this report covers
codegraph_mcp_probe	One of the tools this report covers
resolve1, resolve2, lockfix1, datafrag1	Runner patches covered as kind `runner_patch`
gen_lumenignore, lumen_verify	`scripts/lumen` tools covered
`scripts/codegraph/tests/test_owed_report.py`	The unit tests for this tool

Last verified

2026-09-24T19:08:01Z — ran from `scripts/codegraph`:

- `python3 tests/test_owed_report.py` twice: both `Ran 12 tests ... OK`, exit 0.
- `python3 owed_report.py --strict >/dev/null; echo $?` printed `1` (tools were still owed at that time).
- `python3 owed_report.py --root /nonexistent_zz` printed `REFUSED: root is not a directory: /nonexistent_zz`, exit 3.

- `python3 owed_report.py --root <empty dir>` printed the `no governed indexing tool discovered` refusal, exit 3.
- `python3 owed_report.py --bogus` exit 2.
- `python3 owed_report.py --json` top-level keys: `owed_count`, `root`, `tools`; row keys `guide kind owed path tested tests`.